Computer Science 10 Calendar

Autumn, to the Christmas break · OCR J277

We meet **five times a fortnight** — in **Week A** a Tuesday double (P5 and P6) and Wednesday P3; in **Week B** Tuesday P1 and Thursday P1. Until mid-November we work through the **Hardware** booklet, and the **Hardware test is Wednesday 18 November**. Then we design programs before we write them: **flowcharts and decisions**, with a **design test on Tuesday 15 December**. **Our last lesson before Christmas is Wednesday 16 December.**

Week	Mon	Tue	Wed	Thu	Fri
26 Oct · B	26	**27** P1	28	**29** P1	30
2 Nov · A	2	**3** P5+P6	**4** P3	5	6
9 Nov · B	9	**10** P1	11	**12** P1	13
16 Nov · A	16	**17** P5+P6	**18** TEST	19	20
23 Nov · B	23	**24** P1	25	**26** P1	27
30 Nov · A	30	**1 Dec** P5+P6	**2 Dec** P3	3 Dec	4 Dec
7 Dec · B	7	**8** P1	9	**10** P1	11
14 Dec · A	14	**15** TEST	**16** P3 *last day of term*	17	18

What happens in each lesson

Lesson	Time	What we do
Hardware — how a computer finds, stores and reads binary		
Tue 27 Oct	P1 · 55 min	Why a computer needs memory · RAM and ROM *(holiday reading due)*
Thu 29 Oct	P1 · 55 min	Cache · virtual memory · primary storage practice
Tue 3 Nov	P5+P6 · 105	Secondary storage · magnetic, solid state, optical · the six characteristics · choosing storage for a job
Wed 4 Nov	P3 · 55 min	Inside the CPU · von Neumann architecture · control unit, ALU, cache, registers
Tue 10 Nov	P1 · 55 min	The fetch-execute cycle · PC, MAR, MDR, accumulator
Thu 12 Nov	P1 · 55 min	CPU performance · clock speed, cores, cache size
Tue 17 Nov	P5+P6 · 105	Embedded systems, then revision from the booklet's "Can I …?" pages
Wed 18 Nov	P3 · 55 min	TEST **Hardware test** · the whole unit
Algorithm design — planning a program before writing it		
Tue 24 Nov	P1 · 55 min	Test feedback · what an algorithm is · inputs, processes and outputs
Thu 26 Nov	P1 · 55 min	Flowcharts · start, process, input and output · a sequence of steps
Tue 1 Dec	P5+P6 · 105	Decisions · *if … then* in a flowchart · following a flowchart by hand
Wed 2 Dec	P3 · 55 min	Decisions with more than one path · flowcharts for real-world jobs
Tue 8 Dec	P1 · 55 min	Repeating steps · loops in a flowchart · trace tables
Thu 10 Dec	P1 · 55 min	From flowchart to pseudocode · design practice
Tue 15 Dec	P5+P6 · 105	P5 revision, then P6 TEST **Design test** · flowcharts and decisions, no code
Wed 16 Dec	P3 · 55 min	Test feedback · what comes next: writing the code, in Python